

Biswadeep Chakraborty

biswadeep23@iiserb.ac.in | +91 8132947599
linkedin.com/in/biswc | github.com/biswakim | biswakim.github.io

EDUCATION

Indian Institute of Science Education and Research, Bhopal

B.S. in Computer Science and Engineering

Expected Aug 2027

(Lowest possible grade: 0, Highest possible grade: 10)

Vivekananda Kendra Vidyalaya, Nirjuli

Final Grade: 95.3%

Higher Secondary (CBSE)

2020–2022

(Lowest possible grade: 0%, Highest possible grade: 100%)

Vivekananda Kendra Vidyalaya, Nirjuli

Final Grade: 98%

Secondary (CBSE)

2010–2020

(Ranked 2nd in the state)

(Lowest possible grade: 0%, Highest possible grade: 100%)

RESEARCH EXPERIENCE

Research Intern, Dept. of Computer Science and Information Engineering –

National Chung Cheng University, Taiwan

June 2026 – August 2026

Taiwan Experience Education Program (TEEP)

Project: Multi-Scale Optimal Transport Framework for Heterogeneous Transformer Fusion

- Developed a **multi-scale Optimal Transport framework** for fusing heterogeneous Transformer architectures by formulating model fusion as a **computational graph alignment** problem.
- Designed a macro-to-micro fusion pipeline combining **Fused Gromov-Wasserstein (FGW) transport**, Wasserstein barycentric synthesis, and Sinkhorn-based parameter alignment.
- Addressed architectural depth mismatch between **6-layer and 12-layer Vision Transformers** through Wasserstein barycentric synthesis instead of heuristic identity padding.
- Evaluated the framework on **CIFAR-100**, achieving **49.74% test accuracy** after only **15 fine-tuning epochs**, outperforming both independently trained parent models.

Summer Research Internship — Indian Institute of Technology, Mandi

June 2025 – July 2025

Worked under the guidance of Dr. Dinesh Singh and mentorship of Tarun Saini (Ph.D.) on projects at the intersection of optimization, robotics, and machine learning. Key contributions:

- Developed and evaluated ILP, Greedy, Simulated Annealing, and a novel Multi-Stage algorithm for drone-based surveillance scheduling under patrol-interval constraints.
- Implemented and tested frameworks on real-world maps (Ritsumeikan University, IISER Bhopal), achieving high coverage efficiency and reduced runtime using the Multi-Stage framework.
- Implemented the **Trajectron** model: a graph-based deep generative framework using CVAEs and dynamic spatiotemporal graphs for multi-agent trajectory prediction.
- Evaluated performance on ETH/UCY datasets, obtaining competitive ADE/FDE/NLL metrics and outperforming baselines such as SGAN.

Gained hands-on experience in optimization, deep generative modelling, spatiotemporal graph networks, and scientific experimentation. **Winter**

Research Internship — Indian Institute of Technology, Bhubaneswar

December 2024 – January 2025

Worked in the School of Electrical and Computer Sciences under Dr. Shreya Ghosh and Dr. Abhik Jana on the Microsoft Academic Partnership Grant 2024 project, “Geospatial Knowledge Inference Using LLMs for Enhanced Travel Decision-Making.” Key contributions:

- Developed personalized travel-planning workflows using large language models (LLMs).
- Performed data engineering tasks: web scraping, API integration, and dataset curation using Python (Playwright, Geopy).
- Automated geospatial data processing to support scalable real-world analysis.

Gained hands-on experience in LLM-based decision support, automation, and geospatial intelligence.

PROJECTS

Deep Generative Systems for Indigenous Textile Archiving and Design

January 2026 – Present

Independent project applying deep learning to digitize, classify, and re-envision indigenous tribal handloom motifs of Arunachal Pradesh.

- Designed **CNN/GAN pipelines** for regional motif classification and generative synthesis.
- Developing a **searchable dataset** integrating visual data, metadata, and cultural narratives.
- Prototyping an **AI-assisted pattern-design toolkit** for sustainable textile innovation.
- Applying **neural style-transfer** to fuse traditional motifs with contemporary aesthetics.

Demographic-Agnostic Parkinson’s Disease Detection

Apr 2026 – Present

Python, PyTorch, Librosa

- Designing a fairness-aware deep learning framework for Parkinson’s detection using Mel-spectrogram representations of speech signals
- Developing a tri-branch adversarial CNN with Gradient Reversal Layer (GRL) to disentangle pathological features from demographic attributes
- Engineering an audio preprocessing pipeline using VAD, STFT, Mel-scale transformation, and SpecAugment for robust feature learning
- Evaluating cross-linguistic generalization using multi-source datasets (mPower, PC-GITA, Czech PD)

TECHNICAL SKILLS

Programming: Python, C, SQL

Machine Learning: PyTorch, Scikit-learn, OpenCV

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, NetworkX

Mathematics: Linear Algebra, Probability, Statistics, Optimization, Calculus

Tools: Jupyter Notebook, Mathematica, AutoCAD, Microsoft Office

Web & Automation: Playwright, Geopy, API Integration

AWARDS & ACHIEVEMENTS

Acceptance into New York University — got accepted into the Computer Science & Mathematics Program for the Class of 2026

2022

Dera Natung Award for Academic Excellence — Government of Arunachal Pradesh

2021

State Merit Scholarship — Ranked 1st in the state, Government of Arunachal Pradesh

2019

EXTRACURRICULAR ACTIVITIES

Convener | NEVISTA, North-East Fest of IISER Bhopal

January 2026

Co-founded and served as the first Convener of NEVISTA, a cultural festival celebrating the heritage and diversity of North-East India.

Core Committee Member | IISER Bhopal Quiz Club

December 2024 – Present

Contribute to quiz hosting, content development, and event organization.

Core Committee Member | The Poetry Club, IISER Bhopal

September 2024 – April 2025

Created and edited promotional videos and managed digital media for club events.

Content Creation Team | Animal Welfare Society, IISER Bhopal

February 2024 – August 2024

Produced content and media supporting animal welfare initiatives on campus.

Co-Founder | Be The Voice

July 2020 – December 2021

Co-founded an NGO focused on stray animal feeding, medical treatment, adoption, and abuse reporting in the Itanagar capital region.